

End-to-End Deployment Process

Snowflake + Git + GitHub + GitHub Actions

This is the final recommended deployment flow for your Snowflake Retail Analytics project.

Step 1: Create Git Repository

Create repository:

```
1 snowflake-retail-analytics
```

Repository structure:

```
1 snowflake-retail-analytics
2 |
3 |— .github
4 |   |— workflows
5 |     |— deploy-snowflake-dev.yml
6 |
7 |— data
8 |   |— customers.csv
9 |   |— products.csv
10 |   |— stores.csv
11 |   |— sales.csv
12 |
13 |— phase01_environment
14 |— phase02_security
15 |— phase03_ingestion
16 |— phase04_data_quality
17 |— phase05_transformations
18 |— phase06_dynamic_tables
19 |— phase07_streams
20 |— phase08_tasks
21 |— phase09_semistructured
22 |— phase10_time_travel
23 |— phase11_cloning
24 |— phase12_performance
25 |— phase13_external_tables
26 |— phase14_snowpark
```

```
27 |— phase15_data_sharing
28 |— phase16_cortex_ai
29 |— phase17_monitoring
30 |— phase18_cost_optimization
31 |— phase19_governance
32 |— phase20_capstone
33 |
34 |— requirements.txt
35 |— README.md
```

Step 2: Initialize Git

```
1 git init
2
3 git add .
4
5 git commit -m "Initial Snowflake Retail Analytics Project"
```

Step 3: Create GitHub Repository

```
1 https://github.com/new
```

Repository Name:

```
1 snowflake-retail-analytics
```

Connect local repository:

```
1 git remote add origin https://github.com/<username>/snowflake-
  retail-analytics.git
2
3 git branch -M main
4
5 git push -u origin main
```

Step 4: Generate JWT Key Pair

Git Bash:

```
1 openssl genrsa -out rsa_key.pem 2048
2
```

```
3 openssl pkcs8 -topk8 \  
4 -in rsa_key.pem \  
5 -out rsa_key.p8 \  
6 -nocrypt  
7  
8 openssl rsa \  
9 -in rsa_key.pem \  
10 -pubout \  
11 -out rsa_key.pub
```

Step 5: Configure Snowflake User

Run in Snowflake:

```
1 ALTER USER BHASKAR  
2 SET RSA_PUBLIC_KEY='<PUBLIC_KEY>';
```

Verify:

```
1 DESC USER BHASKAR;
```

Step 6: Create GitHub Secrets

Repository:

```
1 Settings  
2 → Secrets and variables  
3 → Actions
```

Create:

```
1 SNOWFLAKE_ACCOUNT  
2 SNOWFLAKE_USER  
3 SNOWFLAKE_ROLE  
4 SNOWFLAKE_WAREHOUSE  
5 SNOWFLAKE_DATABASE  
6 SNOWFLAKE_PRIVATE_KEY
```

Example:

```
1 SNOWFLAKE_ACCOUNT=jqcqaun-fs47684  
2 SNOWFLAKE_USER=BHASKAR  
3 SNOWFLAKE_ROLE=ACCOUNTADMIN  
4 SNOWFLAKE_WAREHOUSE=RETAIL_WH
```

```
5 SNOWFLAKE_DATABASE=RETAIL_DB
```

Step 7: Create requirements.txt

```
1 snowflake-cli
2 snowflake-snowpark-python
3 pyyaml
```

Step 8: Create GitHub Workflow

File:

```
1 .github/workflows/deploy-snowflake-dev.yml
```

Workflow responsibilities:

```
1 Validate YAML
2 Validate Repository
3 Install Dependencies
4 Configure JWT Authentication
5 Test Connection
6 Execute Phase 1
7 Execute Phase 2
8 ...
9 Execute Phase 20
10 Validate Deployment
```

Step 9: Phase Execution Order

```
1 Phase01 Environment
2 ↓
3 Phase02 Security
4 ↓
5 Phase03 Ingestion
6 ↓
7 Phase04 Data Quality
8 ↓
9 Phase05 Transformations
10 ↓
```

```
11 Phase06 Dynamic Tables
12 ↓
13 Phase07 Streams
14 ↓
15 Phase08 Tasks
16 ↓
17 Phase09 JSON
18 ↓
19 Phase10 Time Travel
20 ↓
21 Phase11 Clone
22 ↓
23 Phase12 Performance
24 ↓
25 Phase13 External Tables (Skipped)
26 ↓
27 Phase14 Snowpark
28 ↓
29 Phase15 Data Sharing
30 ↓
31 Phase16 Cortex AI
32 ↓
33 Phase17 Monitoring
34 ↓
35 Phase18 Resource Monitors
36 ↓
37 Phase19 Governance (Skipped)
38 ↓
39 Phase20 Analytics
```

Step 10: Commit Changes

Whenever files are updated:

```
1 git add .
2
3 git commit -m "Updated Phase Scripts"
4
5 git push origin main
```

Step 11: Run GitHub Actions

Open:

- 1 GitHub Repository
- 2 → Actions
- 3 → Deploy Snowflake Retail Analytics DEV

Click:

- 1 Run workflow

Select:

- 1 main

Click:

- 1 Run workflow

Step 12: Monitor Deployment

GitHub Actions:

- 1 ☒ Checkout
- 2
- 3 ☒ Validate YAML
- 4
- 5 ☒ Install Dependencies
- 6
- 7 ☒ Snowflake Connection
- 8
- 9 ☒ Phase01
- 10
- 11 ☒ Phase02
- 12
- 13 ☒ Phase03
- 14
- 15 ...
- 16
- 17 ☒ Phase20
- 18
- 19 ☒ Deployment Complete

Step 13: Validate in Snowflake

```
1 SHOW DATABASES;
1 SHOW SCHEMAS IN DATABASE RETAIL_DB;
1 SHOW TABLES IN SCHEMA RETAIL_DB.RAW;
1 SHOW TABLES IN SCHEMA RETAIL_DB.CURATED;
1 SHOW VIEWS IN SCHEMA RETAIL_DB.ANALYTICS;
```

Step 14: Daily Development Process

Make SQL changes:

```
1 git add .
2
3 git commit -m "Phase 06 enhancement"
4
5 git push origin main
```

Run workflow:

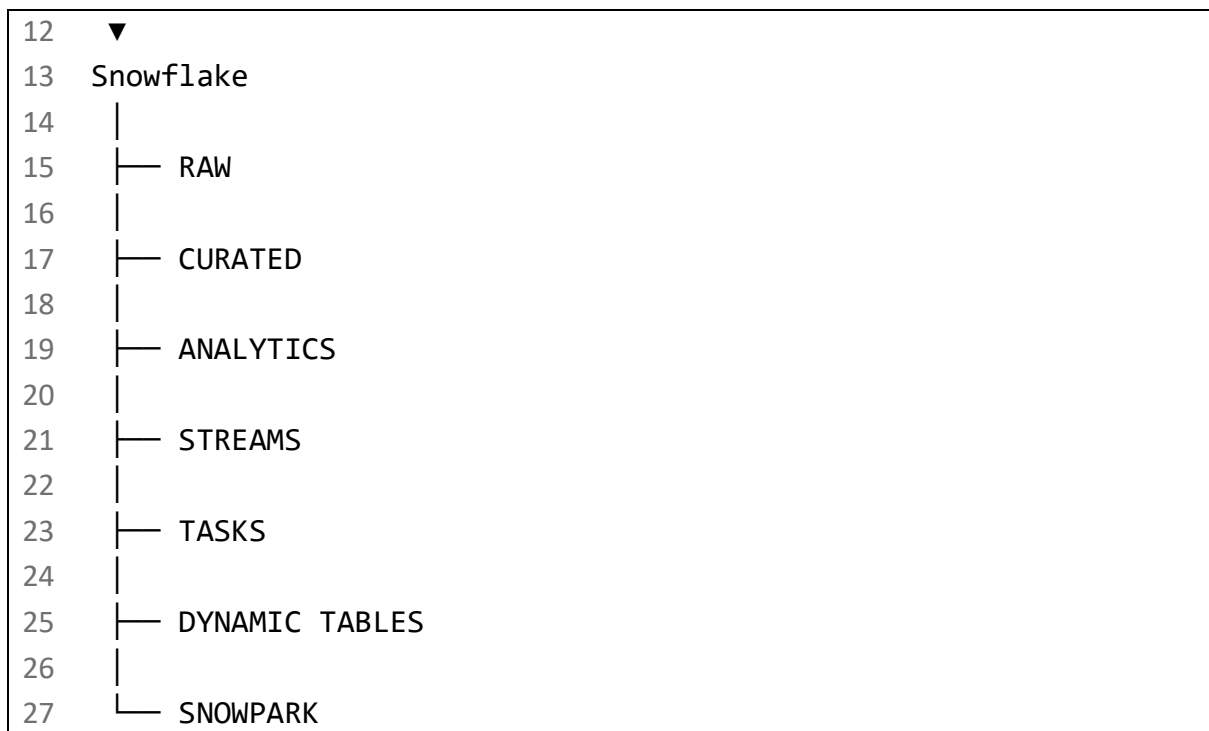
```
1 Actions
2 → Run workflow
```

Verify:

```
1 SHOW TASKS;
2 SHOW STREAMS;
3 SELECT * FROM ANALYTICS.VW_TOP_PRODUCTS;
```

Final Architecture

```
1 Git
2 |
3 ▼
4 GitHub
5 |
6 ▼
7 GitHub Actions
8 |
9 ▼
10 JWT Authentication
11 |
```



Final Recommendation

Use:

- ☒ JWT Authentication
- ☒ Internal Stages
- ☒ CREATE IF NOT EXISTS
- ☒ CREATE OR REPLACE
- ☒ Fully Qualified Names (RETAIL_DB.RAW.SALES)
- ☒ GitHub Actions CI/CD
- ☒ Snowpark Analytics

Skip:

- ☒ External Tables (no cloud storage)
- ☒ Materialized Views (unsupported edition)
- ☒ Masking Policies (unsupported edition)

This gives you a stable, repeatable Snowflake portfolio project and CI/CD pipeline.